

Week 1 Practice: Setup & First Predictions

AI Frameworks

Practice goals (80 minutes)

By the end of this session you should have:

1. A working environment (Colab *or* local Jupyter)
2. A notebook where you can run cells and see output
3. A simple plot from a tiny list of numbers
4. A hand-made rule that “predicts” a house price from size

Agenda

1. Environment check (15–20 min)
2. Notebook warm-up (15 min)
3. First tiny plot (15 min)
4. Predict-by-hand activity (20–25 min)
5. Done checklist & share-out (10 min)

Option A — Google Colab (recommended if unsure)

1. Open `https://colab.research.google.com`
2. Sign in with a school or personal Google account
3. File → New notebook
4. Run a cell: `print(1+1)` — you should see 2

No local install needed; runs in the browser.

Option B — Local Jupyter

- If the instructor prepared Anaconda / VS Code: follow the handout
- Create a folder for this course
- Launch Jupyter Lab or Notebook; create `week01.ipynb`

Common mistake

Spending the whole session only on installers. If stuck after 15 minutes, switch to Colab and continue learning.

Checkpoint A — verify Python

```
1 import sys
2 print(sys.version)
3
4 import matplotlib
5 print("matplotlib OK")
```

If both print without errors, you are ready.

How a notebook works

- **Code cells:** run Python
- **Markdown cells:** notes and headings (optional today)
- Run order matters: if a cell uses a variable, define it first

Try this

Restart the kernel once today on purpose, then re-run cells from the top.

Warm-up cells

```
1 print("Hello, AI Frameworks!")
2
3 name = "Alex"    # change me
4 print("Student:", name)
5
6 sizes = [50, 65, 80, 95]
7 print("Number of houses:", len(sizes))
8 print("First size:", sizes[0])
```

A minimal plot

```
1 import matplotlib.pyplot as plt
2
3 sizes = [50, 65, 80, 95]
4 prices = [180, 200, 210, 250] # made-up
5
6 plt.scatter(sizes, prices)
7 plt.xlabel("Size (m^2)")
8 plt.ylabel("Price (k)")
9 plt.title("Tiny housing example")
10 plt.show()
```

What to notice

- Larger size tends to go with higher price *in this toy data*
- Real data will be messier—that is normal
- Plots help you think before you model

Activity: invent a rule

Goal: predict **price** from **size** only—no ML library.

1. Look at the scatter plot (or the table of pairs)
2. Invent a simple rule, e.g. $\text{price} = a * \text{size} + b$
3. Pick numbers for a and b that feel reasonable
4. Predict for a *new* size (e.g. 70 m^2) not in the list

Template

```
1 def predict_price(size):  
2     a = 2.0      # tweak these  
3     b = 50  
4     return a * size + b  
5  
6 print(predict_price(70))  
7  
8 # Optional: compare your rule to the toy points  
9 for s, p in zip(sizes, prices):  
10     print(s, "real:", p, "my guess:", predict_price(s))
```

Discuss with a neighbour (3 minutes)

- Whose rule fits the points better “by eye”?
- What happens if you change a a lot?
- Why might two people choose different rules?

Next weeks: algorithms will search for good rules using data—same idea, more systematic.

Quiz

In supervised learning terms, house **size** is a ... and **price** is a ...

1. label ; feature
2. feature ; label
3. library ; notebook
4. plot ; colour

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

2. Size is a **feature** (input); price is the **label** / target.

Done checklist

Tick when finished:

1. Environment works (Colab or local)
2. Notebook runs `print` and imports `matplotlib`
3. Scatter plot of size vs price appears
4. `predict_price` function returns a number for size 70
5. You can say in one sentence what a feature and a label are

Before you leave

- Save / download your notebook (`week01.ipynb`)
- Note one thing that was confusing—bring it to Week 2

Next week

Lecture: Python enough for data (lists, dicts, loops, functions)

Practice: small coding warm-ups without pandas (yet)